Examiner reports

Q1.

The different format of this question took many students by surprise. Many weaker students were confused about the processes required. Some even used integration in part (a) and gradients in part (b) or thought that they were required to find the normal and the tangent in the two parts.

(a) The majority realised the need to substitute x = 1 into the given expression for $\frac{dy}{dx}$, but even that was not always evaluated correctly. Quite a few differentiated the given expression and scored no marks at all. Although most went on to find the equation of the tangent in the required form, a substantial number found the normal instead. Others hedged their bets by finding the equations of both the tangent and the normal, with no indication as to which was which. It was disappointing to see that several students, who obtained a correct equation for the tangent in the form $y - 4 = 9(x - 1)$, were unable to simplify this correctly to the form $y = 9x - 5$.

(b) When integration was attempted, it was usually done correctly with just the odd mistake in signs or coefficients. However, many failed to include the constant of integration, " + c" , and could proceed no further. Many who did have the " + c" stopped at that point. Some substituted x = 1 into their expression for x and equated this to zero. This could not be given any credit because they did not have an expression of the form y = …. and so were unable to substitute y = 4. Many of those who found the correct value of the arbitrary constant failed to write the final equation in the form y = f(x).

Q2.

The use of the trapezium rule was well understood with a majority of students presenting a fully correct solution to part (a) with the final answer given to the degree of accuracy requested. Students need to take extra care when transferring correct decimal values from a table into the formula; this was particularly relevant in this series to those values which had leading zeros, namely for f(4) and f(5). The most common method error was to use end values 0 and 4 for x instead of the correct end values 1 and 5. A small minority of students left their final answer in the form $\frac{694}{1105}$ which resulted in the loss of the final accuracy mark. In part (b)(i), most students were able to integrate one of the terms correctly but it was not uncommon to see the integral of $x^{-\frac{3}{2}}$ resulting in a positive coefficient or an incorrect numerical simplification of 1/–0.5 . Very few students failed to score the method mark in part (b)(ii) for being able to deal with limits in a correct manner.

Q3.

This question tested expansion of brackets and integration of negative exponents.

The majority of students used the binomial expansion or expanding brackets successfully in part (a). Typically in a(ii) they repeated their working from part (a)(i), replacing y with x^{-2} , rather than replacing it in their simplified expansion. Most students scored at least M1 for this part, however a significant number of students were unsuccessful in finding a correct expression for $(2 - x^{-2})^3$ and hence did not find the correct values of p and q.

Those students who did not use “hence” but used a correct different method to the one they used in a(i), could gain a maximum of 2 out of the 3 marks. Part (b) made use of the answer to a(ii) and those that had found values for p and q were generally able to correctly integrate their expression, $p + qx^{-4}$, and evaluate it using the given limits in a correct manner.

Q4.

This final question proved to be less demanding than has been the case in many of the past papers for this unit. Even the unstructured final part of the question produced a significant number of fully correct solutions from the candidates. Parts (a) and (b)(i) were found to be very straightforward with most candidates scoring heavily. It is worth recording that a number of candidates in part (b)(ii) failed to give sufficient justification for taking the gradient m to be –8. With the answer given, it was not acceptable to just state that the gradient is –8 without some clear indication that this value came from the value of $\frac{dy}{dx}$ when x = 8. In part (c), the most common loss of a mark was through failing to simplify $\frac{3}{\left(\frac{8}{3}\right)}$ correctly. The error of integrating $x^{\frac{5}{3}}$ to get a term in the form $kx^{\frac{7}{3}}$ was penalised more heavily.

In part (d), those who found the area of the triangle by solving the equations of OP and AP simultaneously to find the coordinates of P and then using $A = \frac{1}{2}bh$ were generally much more successful than those who attempted to use integration, as their limits were frequently incorrect: 0 to 8 in both cases rather than 0 to 3.2 for OP and then 3.2 to 8 for AP. It was also surprising to see a significant minority of candidates finding the lengths of OP and AP and the angle OPA and then using $\frac{1}{2}ab\sin C$ to find the area of the triangle; although a valid approach it took much more time and usually led to rounding errors. The method for finding the area under the curve was well understood but errors in part (c) frequently resulted in the area under the curve being greater than the area of the triangle: a clear error which should have been spotted by reference to the diagram.

Q5.

In part (a)(i), the coefficient of x, being an odd number, caused problems for some when completing the square; most students could not square 1.5 without a calculator; those who used fractions could not always simplify $5 - \frac{9}{4}$, and for those using decimals –1.25 + 5 = 3.75 was commonly seen.

In part (a)(ii), the equation of the line of symmetry was not usually stated correctly, with several writing down the equation of a curve rather than a straight line; the incorrect equation $y = \frac{3}{2}$ was seen far too often, whereas others simply wrote down numbers such as $\frac{3}{2}$ or $\frac{11}{4}$; this topic was not well understood by many students.

In part (b)(i), most attempts earned the method mark for eliminating y and collecting like terms, but, surprisingly, not all were able to solve the resulting quadratic equation. A common incorrect solution to $x^2 - 4x = 0$ was $x = \pm 2$, whereas others obtained $y = x^2 - 4x$ from the two equations and made no progress.

Integration of polynomials is well drilled and part (b)(ii) earned full marks for weak and strong students alike. There were few errors seen but the most common involved the constant term; some kept the term as 5 and others omitted the final term possibly confusing differentiation with integration.

In part (b)(iii), most students realised the need to substitute appropriate limits but, without a calculator, arithmetic errors were abundant; 4^3 was often evaluated as 48 and the fractions and minus sign caused problems for many students who could not simplify $\frac{64}{3} - 24 + 20$.
Those who were unable to complete part (b)(i) often chose the value 5 as their upper limit, possibly since the coordinates of A were (0, 5), but no credit was given for this. However, those who had the solution $x = 2$ from their earlier work were given a method mark for using this value as their upper limit. A large number of students presented the value of the definite integral as their final answer to the area of the shaded region instead of considering the difference of the area of the appropriate trapezium and the value of the integral.

Q6.

This question, which tested positive rational exponents and integration, was another good source of marks for many students, although for a large minority finding the square of $x^{\frac{3}{2}}$ proved to be more of a challenge than was hoped, with $x^{\frac{9}{4}}$ appearing far too often from otherwise promising students. The indefinite integration and evaluation of a corresponding definite integral were tackled with much more success, with many correct solutions seen.

Q7.

Most candidates replaced $\frac{8}{x^4}$ by $8x^{-4}$ and differentiated correctly to score full marks for part (a).

Some other candidates failed to score the accuracy mark because they included a '$+c$' in their

answer. A very large majority of candidates realised that the value of $\frac{dy}{dx}$ at $x = 1$ was required in
part (b), but a small minority did not appreciated that the value of y was also required. Although many candidates found the equation of the tan gent correctly, it was not uncommon to see others finding the equation of the normal instead.

In part (c), a majority of candidates equated their expression for $\frac{dy}{dx}$ to zero and attempted to
solve the resulting equation, although only about half of the candidates obtained the correct coordinates of M. Those who had wrong powers of x in their answer to part (a) were restricted to just two marks in part (c). The most common incorrect methods seen either involved solving
$\frac{d^2y}{dx^2} > 0$ or looking for a link between the tangent in part (b) and the stationary point.

In part (d)(i), a high percentage of candidates were able to integrate $\frac{8}{x^4}$ correctly, but errors in
the integration of $(x + 3)$ or the omission of the constant of integration resulted in a significant proportion of these candidates failing to score the final mark. Since answers to part (d)(ii) required explicit evidence that the answer to part (d)(i) had been used; the few candidates who just wrote down the answer without any other working gained no credit. However, most candidates applied the 'Hence', showing the values 2 and 1 substituted into their answer to part (d)(i) and carrying out the relevant subtraction. However, a significant number of candidates made arithmetical errors in carrying out the subtraction and so did not score the accuracy mark.

Only the more able candidates appreciated what was required in the final part of the question. The common wrong answers included 0 (presumably from the x-axis being a tan gent), 12 (the y value from part (b)) and 5.5 (forgetting to consider the direction of the translation).

Q8.

The majority of candidates answered part (a) successfully using the binomial expansion, Pascal's triangle or expanding brackets. Those candidates who wrote the expansion using the binomial coefficient notation from the formulae book did not always convince the examiner that they knew how to correctly evaluate the combinatorial coefficients.

In part (b)(i) very few candidates scored full marks with a large number of attempts seen where the numerator and denominator were integrated separately. Those candidates who realized the need for negative indices generally scored well. Most candidates picked up the method mark in the final part of the question for dealing with the limits in a correct manner. A few candidates just stated the correct answer for the definite integral without showing any working. Such an approach gained no credit as it did not explicitly indicate the use of 'Hence'.

Q9.

The sign of p was often awry, although that of q was usually correct. This was not penalised again in part (b)(i) where a lack of simplifying the coefficients and signs in the expressions was a greater source of mark loss. It was also disappointing to see a significant minority of candidates integrating the term + $\frac{1}{x^6}$ instead of the printed term –

$\frac{1}{x^6}$. However, in general, most candidates showed that they understood how to integrate x raised to a negative power.

Most knew how to evaluate a definite integral in part (b)(ii) but very few could maintain the numerical accuracy needed for the final mark. Those few candidates who just wrote down the value of the definite integral, presumably from a calculator, without showing how it had been obtained from the previous part (Hence) were awarded no credit. Some candidates, having correctly obtained '–1.7' for the value of the definite integral, went on to write 'Area = 1.7' as their final line of solution. This was not penalised and such candidates were awarded both marks.

Q10.

In part (a)(i) the majority of candidates realised the need to find the value of f(x) when x = 2. However, it was also necessary, after showing that f (2) = 0, to write a statement that the zero value implied that x – 2 was a factor. It was good to see more candidates being aware of this.

In part (a)(ii), those who used inspection were the most successful here. Methods involving long division or equating coefficients usually contained algebraic errors.

In part (b)(i) a surprising number of candidates failed to see the need to differentiate in order to find the gradient at Q. Those who attempted to find $\frac{dy}{dx}$ sometimes wrote it as $3x^2 + x$, but usually were aware of the need to substitute x = 2.

In part (b)(ii) those who had the correct gradient in part (b)(i) were usually successful in finding the correct equation of the tangent, and most obtained at least a method mark here.

In part (b)(iii) most were well drilled in integration and earned full marks, although some wrote $\frac{x^4}{4} + \frac{x^2}{2} - 10$ and others gave $\frac{x^4}{4} + \frac{x^2}{2} - \frac{10^2}{2}$ as their answer. For part (b)(iv) the correct limits were usually used, although many sign/arithmetic slips occurred after substitution of the numbers 0 and 2 and it was incredible how many could not evaluate 6 – 20 without a calculator. Very few candidates realised the need to show clearly that, although the integral from 0 to 2 gave a value of –14, the area of the shaded region was 14. A separate statement was needed and those who simply wrote 4 + 2 – 20 = –14 = 14 did not score full marks. Those more able candidates who made a statement about the region being entirely below the x-axis and who subsequently evaluated the integral from 2 to 0 correctly scored full marks.

Q11.

Most candidates gave the correct value for n in part (a). The most common wrong answer was '$n = -\frac{1}{4}$'.

The expansion of $\left(1+\frac{3}{x^2}\right)^2$ in part (b) caused more difficulty than expected. A common wrong answer was '$1 + 6x^{-2} + 9x^4$' although there were many other examples of incorrect answers offered.

In integrating $\left(1+\frac{3}{x^2}\right)^2$ in part (c) most candidates realised that their expansion in part (b) should be used but surprisingly the integration of 1 with respect to x caused just as many problems as the integration of the terms in x raised to negative powers. Although many candidates showed that they knew how to deal with the given limits, a lack of brackets in their numerical expressions frequently led to the wrong answer. The final mark was frequently lost because no **exact** answer (fraction or mixed number) was given.

Q12.

Generally, most candidates scored relatively high marks for this long structured calculus question. Parts (a)(i), (ii) and (iii) were answered very well, although some used the gradient of the normal in part (a)(iii) as –2 and so in part (a)(iv) obtained a positive value for the y-coordinate of Q. A glance at the diagram should have indicated that such a value was incorrect.

Part (a)(v), which required candidates to find the x-coordinate of the maximum point, M, was generally not answered with any confidence. To score any marks for this part, candidates were required to go beyond just equating their answer for part (a)(i) to zero. Many candidates again incorrectly believed that for a maximum point the second derivative was zero. Candidates generally found the correct answer to the indefinite integral in part (b)(i), but a significant minority forgot to add the area of the triangle to their 'area under the curve' in part (b)(ii) of the question.

Q13.

Although many candidates applied a correct method to find the expansion of $\left(1+\frac{4}{x^2}\right)^3$, accuracy marks were sometimes lost because brackets were missing and the factor 4 was not squared when finding the value of q. Although a significant minority of candidates failed to use the expansion in part (a), despite the word 'Hence', most other candidates obtained the method mark in part (b), but it was surprising to find candidates having more difficulty integrating '1' than integrating the terms in x. Many candidates scored the method mark in part (b)(ii) but arithmetical errors were common in the subsequent evaluation.

Q14.

Part (a)(i) Some candidates ignored the request to state the coordinates of B even though they were using the height of the triangle as 5. The negative x-coordinate of A caused quite a few to feel that the triangle had a negative area. Far too many when finding $\frac{1}{2}\times 1\times 5$ wrote the answer as 3.

Part (a)(ii) Practically every candidate found the correct integral although some made errors when cancelling fractions.

Part (a)(iii) It was necessary here to have the lower limit as –1 and the upper limit as 0. Many reversed the order and by some trickery arrived at a positive value. This was penalised and so very few, even though many had an answer of 1 for the area, scored full marks for this part of the question.

Part (b) Most candidates differentiated correctly but, because of poor understanding of negative signs, many wrong values of −13 were seen for the gradient. There is obviously confusion for many between tangents and normals and several thought the gradient of the tangent was $\frac{-1}{17}$.

Q15.

This calculus question again proved to be a good source of marks for many candidates.

Part (a)(i) was answered very well but in part (a)(ii) a significant minority used $\frac{d^2x}{dx^2} = 0$ instead of $\frac{dy}{dx} = 0$ dx^2. In part (a)(iii), although most candidates showed that they understood the method to find the gradient of the tangent, a significant proportion failed to find and use the gradient of the normal. Many candidates showed that they had a good understanding of integration but, surprisingly, errors in integrating (x + 1) were almost as common as errors in integrating $\frac{4}{x^2}$. Examiners expected to see some simplification of the algebraic term $\frac{+4x^{-1}}{-1}$. In the final part, the evaluation of F(4) − F(1) as 8.5 was the most common error.

Q16.

Part (a)(i) was well answered, with the main error being $\int -e^x\,dx = -2e^{2x}$. In part (a)(ii) the answer was given. Frequently candidates tried to ‘fudge’ their solutions to arrive at the correct result, with common error being $4\ln 2 - 2 - \frac{1}{2} = 4\ln 2 - \frac{3}{2}$, the given answer.

Part (b)(i) was again well answered by the majority of candidate, although in part (b)(ii) the step from 2x = ln 4 to obtain x = ln 2 was often unconvincing. Some candidates answered part (c) very well and gained full credit. The main error was using 8 or –8 as the gradient of the normal, however many candidates earned at least partial marks.

Although correct answers were seen in part (d), they were not common. Many candidates earned the method mark by the substitution of x = 0 and some went on to find the area of the triangle. The final mark was often lost through a failure to realise that the area was positive.

Q17.

Part (a)(i) Most candidates realised the need to find the value of f(x) when x = 1. However, it was also necessary, after showing that f(1) = 0, to write a statement that the zero value implied that x – 1 was a factor.

Part (a)(ii) Those who used inspection were the most successful here. Methods involving long division or equating coefficients usually contained algebraic errors.

Part (a)(iii) This section seemed unclear to some candidates. Many tried to find the discriminant but used the coefficients of the cubic equation. Many who used the quadratic thought that in order to have one real root the discriminant had to be zero, no doubt thinking the question was asking about equal roots. Some correctly stated that 1 was the only real root but many were obviously confused by the terms "factor" and "root" and stated that "x – 1 was a root".

Part (b)(i) Most candidates were well versed in integration and earned full marks here.

Part (b)(ii) The correct limits were usually used, although many sign/arithmetic slips occurred after substitution of the numbers 1 and 2. Very few candidates realised the need to find the area of a triangle as well and so failed to subtract the value of the integral from the area of the triangle in order to find the area of the shaded region.

Q18.

Most candidates were able to correctly simplify the given expressions although $x^{\frac{9}{4}}$ was a common wrong answer in part (a)(iii) for the simplification of $\left(x^{\frac{3}{2}}\right)^2$. The majority of candidates displayed a good understanding of integration but a significant proportion of the entry left their answer to part (b)(i) in the form $\frac{3}{\frac{3}{2}}x^{\frac{3}{2}}$. The final mark of three required the answer to be simplified further and the constant of integration present. The majority of candidates showed sufficient working in part (b)(ii) to show that they had used their answer to part (b)(i) and so gained the method mark, but a significant minority failed to evaluate 2× $9^{\frac{3}{2}}$ correctly.

Q19.

(a) Not everyone recognised that the height of the rectangle was the value of y when x = –1 or x = 4. Some who did made numerical errors. Even having found the height 4, some obtained the wrong area by taking AB as 4 or 2 (sometimes using Pythagoras' Theorem) or by finding the perimeter instead.

(b) (i) The integral was generally correct, though sometimes incorrect simplification occurred subsequently. A few integrated x^3 to $\frac{1}{4}x^4$ and a surprising number misread the integrand as $3x^2 - x^2$. There were also candidates who confused integration with differentiation or whose process was a hybrid of the two.

(ii) Almost everyone recognised that they should firstly evaluate the integral from –1 to 2, but most stopped there, instead of going on to subtract the value of the integral from the area of the rectangle. There were a lot of sign errors in the work with some adding instead of subtracting or putting the two parts the wrong way round. A few wrongly substituted in the original function. Those who chose to work with the differences of two integrals seldom completed it correctly.

(c) (i) Differentiation was done well on the whole.

(ii) Many substituted $x = 1$ into the derivative to find the gradient of the tangent and went no further. Most did not find the y coordinate of the point and so made no attempt at the equation of the tangent. A few non-linear equations were seen with a 'gradient' of $6x - 3x^2$.

(iii) Very few candidates completed this part. Many made no attempt, and those who did tended to test a few values of x or to find the second derivative, which was of no value.

Only the strongest candidates realised that $\frac{dy}{dx} < 0$ was the condition for y to be decreasing and that, after a couple of lines of algebra, the given inequality could be obtained.

(d) Although most made an attempt at the quadratic inequality, few obtained both parts of the solution. It was imperative that candidates wrote $x > 2$, $x < 0$ and not $0 > x > 2$ as many incorrectly stated.

It was disappointing to see how many candidates at this level could not solve the equation $x^2 - 2x = 0$, obtaining values such as –2, √2, 1+√2. Using the formula or completing the square sometimes led to 1 ± √1, which many candidates failed to simplify.

Q20.

This final question was a good source of marks for many candidates. Even the weaker candidates were able to give correct answers for parts (a) and (c) which tested standard C2 work on differentiation and integration. A significant minority of candidates started part (b)(i) by equating dy/dx to zero instead of substituting $x = 0$ to find the value of the derivative. Those who were able to answer the first two parts of part (b) correctly generally went on to find the correct coordinates of P. A significant number of candidates left their final answer as 24.3 without considering the area of the triangle OAP and carrying out the subtraction. For those candidates who attempted to find the area of the triangle some applied very lengthy methods which involved finding the length of each side of the triangle then using the cosine rule to find angle OPA then using the formula $\frac{1}{2}\ ab \sin C$ or involved the integration of the equations of the two

lines. To their credit a number of such candidates did avoid premature approximation to reach the correct value for the area of the triangle which could have been obtained more directly by using $\frac{1}{2}OA \times |y_P|$.

Q21.

Part (a)(i) Most candidates differentiated correctly. However a few made a slip or misread one of the terms.

Part (a)(ii) Although most managed to substitute 2 into their derivative some made numerical errors and some used y or the second derivative. Most who realised that they should equate their derivative to zero (or at least showed their intention though never inserting the = 0) then tried to factorise or use the formula (though it was clear that some did not recognise the quadratic equation as such). It was disappointing that 14 the bracket $(3x - 14)$ often produced the solution $x = 14$ instead of 14/3, and the solution of $x = 2$ did not always appear.

Part (b)(i) The integration was also completed correctly by most candidates, although the $28x$ was occasionally wrong and some 'hybrid' processes led to terms such as $\frac{20x^3}{3}$

In part (b)(ii) almost everyone attempted to substitute 3 into their integral but their problems with the ensuing fractions often took pages to resolve, and although most ended 'magically' with the required answer there were many errors en route. A few substituted into the original expression for y instead of the integrated expression.

Part (b)(iii) This part was quite well done although again there were errors in evaluating both $\frac{1}{2} \times 21 \times 3$ and $56\frac{1}{4} - 31\frac{1}{2}$. Some candidates confused length with area and merely used Pythagoras's Theorem to find the length of the hypotenuse of the triangle. Those who chose to integrate the equation of the straight line were sometimes successful but many made arithmetic errors.

Q22.

Many candidates lost marks on this question from careless work or failure to write answers to the correct degree of accuracy.

Part (a) Although many candidates obtained the correct method mark $\frac{dy}{dx} = x \times \frac{1}{x} + \ln x$, a significant number who reached this stage were unable to cope with $x \times \frac{1}{x}$ and equated it to 0, giving an answer of 1n x.

Part (b) Many candidates failed to realise the connection with the previous part. A significant number of candidates just wrote down the correct answer with no method indicated.

Part (c) Many candidates scored the method mark; some for the correct substitution and many for following through their answer to (b). Some candidates lost the accuracy mark for not writing the answer in exact form but going straight from (5ln5 – 5) – (1ln1 – 1) to

4.047. Surprisingly, a number of candidates could not evaluate –5 – (–1) correctly.

Q23.

(a) (i) Some were confused by the word "factor" and gave answers as 1 and –2.

(ii) Although there were many correct answers given for the factorisation, some candidates appeared to be guessing the factors of f(x), making no use of the information given in part (a)(ii). Despite getting a non-zero remainder for part (a)(i), several candidates gave (x + 1) as a factor.

(b) Most candidates found the coordinates of A, and those with the correct factor (x – 4) of f(x) were usually able to find the correct coordinates of B.

(c) It was pleasing to see most candidates able to integrate correctly, although quite a few failed to integrate 8 to give 8x. Some candidates did not evaluate the limits in the correct order, but there were many correct values seen for the area of the shaded region.

Q24.

Parts (a) and (b) were answered very well and although the powers in answers to part (c) were nearly always correct, the coefficients were sometimes the reciprocal of the powers in part (b). Part (d) caused significantly more problems to the candidates. Although the method of solving a definite integral was well understood, many candidates resorted to using decimal approximations and, in general, candidates displayed a weakness in converting fractional powers of numbers back into surds.

Q25.

This question was answered very well by many candidates, and evidently the basic processes of differentiation and integration had been well rehearsed.

Part (a)(i) Most candidates substituted x = 3 into the given cubic equation and found p(x) to be zero. Some, however, lost a mark as they did not indicate that p(3) = 0 implied that x – 3 was a factor.

Part (a)(ii) The coordinates (3, 0) were often seen but some forgot to give the y coordinate

Part (b)(i) Almost all candidates differentiated correctly, which was pleasing.

Part (b)(ii) Most candidates realised the need to equate dy/dx to zero. However not all were able to solve the resulting equation. Some of those who did solve the quadratic offered both solutions, 1 **and** 7/3. Many found the second derivative in order to ascertain which point was the minimum point M. This was unnecessary as the coordinates of A (1, 0) were clear from the diagram.

Part (c) The second derivative was almost always found correctly followed by the substitution of x = 1.

Part (d)(i) The integration was carried out very well. Occasionally $\frac{3x^4}{4}$ was seen but this was rare.

Part (d)(ii) Most candidates used the correct limits of 0 and 1. However many arithmetic errors occurred.

When trying to combine the fractions so that the correct value of the integral $-\frac{11}{12}$ was not seen very often. Of those who did integrate correctly, many still lost the last mark for not stating that the area of the shaded region was 11/12 and the negative value of the integral arose because the region was below the x-axis. It was not acceptable to simply write Area = $-\frac{11}{12}=\frac{11}{12}$.

Q26.

This question proved to be a good source of marks for many candidates. In part (a)(i) most candidates quoted the correct value for the power p. Although many candidates produced correct answers for the integration, all the usual errors, as illustrated by $\frac{3}{2}x^{1.5}, x^{1.5}, \frac{x^{-0.5}}{-0.5}$, and $\frac{\sqrt{x}^2}{2}$, were seen. Most candidates stated and used the correct limits in part (a)(iii) but evaluation was sometimes poor. Many of the candidates found a correct equation for the tangent but it was surprising to find a significant minority of these candidates either making no attempt to find the area of triangle AOB or giving the area as a negative value. For full marks in part (c) the examiners were expecting to see the word ‘translation’. Some candidates wrote ‘tr’ but this was not considered to be clear enough to distinguish between ‘transformation’ and ‘translation’. Common wrong answers involved ‘stretch’ and vectors in the wrong direction. In part (d) there was a significant minority of candidates starting their solution with the incorrect statement $\sqrt{x-1}=\sqrt{x}-\sqrt{1}=\sqrt{x}-1$ but in general the trapezium rule was well understood and many candidates were able to give the final answer to the correct required degree of accuracy in this numerical integration question.

Q27.

Part (a) was reasonably well answered although there were some candidates who integrated using the product rule. In part (b)(i) candidates were expected to use the chain rule, but some candidates tried to expand the brackets. This method could score full marks but candidates who used this approach, in general, made mistakes on the expansion. Part (b)(ii) used the previous part; however a large number of candidates tried to use integration by parts and found this difficult. Candidates should be guided to look for relationships between parts of a question.